

dtox helps AI agents test Web3 ideas against scientific research.

Source-linked papers, equations, and author-stated limits.

BEV AUCTION MODEL / EQUATION 4

$$b_i^{\text{PA}}(\mathcal{O}, \mathcal{S}_i) = \frac{n-1}{n} \mathcal{R}_i(\mathcal{O})$$

Equation + assumptions

**Quantifying Blockchain
Extractable Value**

arXiv:2101.05511

A relevant paper can hide a critical assumption.

SOURCE TEXT / BEV PAPER

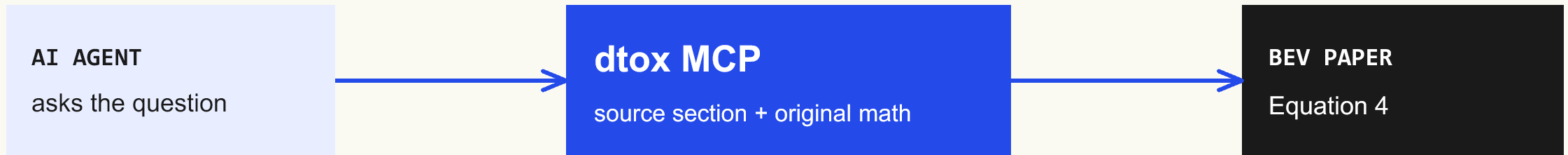
independently drawn from the same uniform distribution, i.e.,

In this model, independent player rewards share one known uniform distribution. The participant count n is known.

DESIGN IMPLICATION

Check whether your protocol's rewards fit that model before applying its bid.

What bidding strategy does the BEV auction model derive, and under which assumptions?



DERIVED BID / EQUATION 4

$$b_i^{\text{PA}}(\mathcal{O}, \mathcal{S}_i) = \frac{n-1}{n} \mathcal{R}_i(\mathcal{O})$$

arXiv:2101.05511 / Exact LaTeX
retained in source notes

The bid scales the player's reward by $(n-1)/n$.

The original bid rule

$$b_i^{\text{PA}}(\mathcal{O}, \mathcal{S}_i) = \frac{n-1}{n} \mathcal{R}_i(\mathcal{O}) \quad (4)$$

EXACT LATEX / `get_code_or_math_spec`

```
b_i^{\text{PA}}(\mathcal{O}, \mathcal{S}_i)=\frac{n-1}{n}\mathcal{R}_i(\mathcal{O})
```

ASSUMPTION IN THE SAME SOURCE

independently drawn from the same uniform distribution, i.e.,

Independent rewards, common uniform distribution, known n

Authors' limits, before you build

Security Considerations

The `EntryPoint` contract will need to be audited and formally verified, because it will serve as a central trust point for *all* [ERC-4337]. In total, this architecture reduces auditing and formal verification load for the ecosystem, because the amount of work that individual *accounts* have to do becomes much smaller (they need only verify the `validateUserOp` function and its “check signature and pay fees” logic) and check that other functions are `msg.sender == ENTRY_POINT` gated (perhaps also allowing `msg.sender == self`), but it is nevertheless the case that this is done precisely by concentrating security risk in the `EntryPoint` contract that needs to be verified to be very robust.

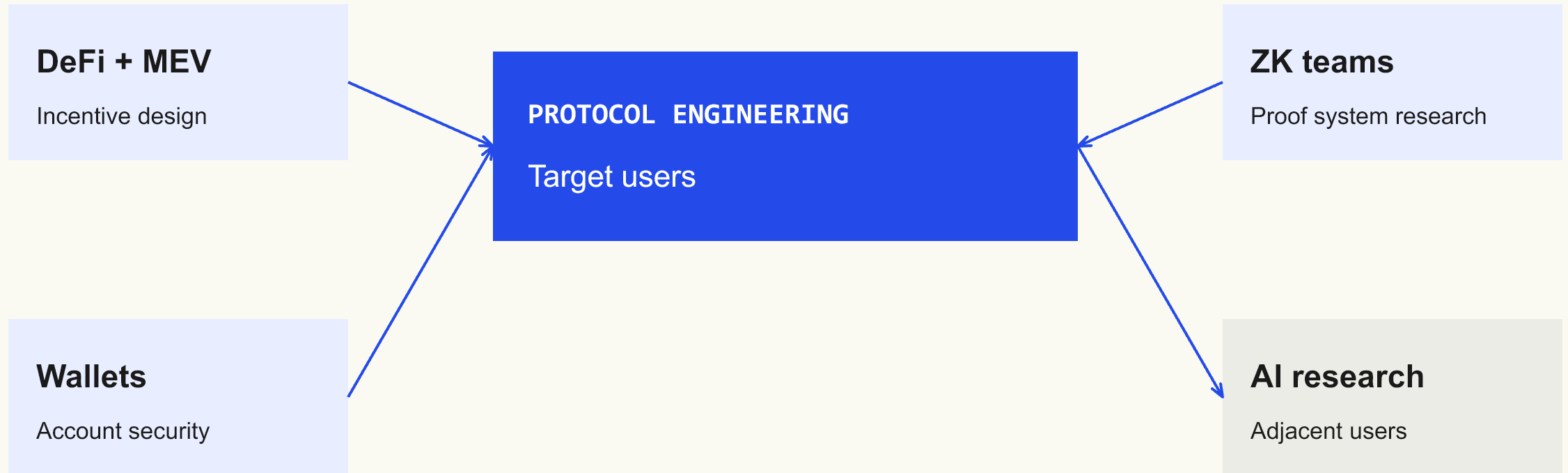
Verification would need to cover two primary claims (not including claims needed to protect paymasters, and claims needed to establish p2p-level DoS resistance):

- **Safety against arbitrary hijacking:** The `EntryPoint` only calls to the `sender` with `userOp.calldata` and only if `validateUserOp` to that specific `sender` has passed.
- **Safety against fee draining:** If the `EntryPoint` calls `validateUserOp` and passes, it also must make the generic call with calldata equal to `userOp.calldata`

EntryPoint becomes a central trust point. Check it before choosing an architecture.

INITIAL CUSTOMER

Protocol engineering teams

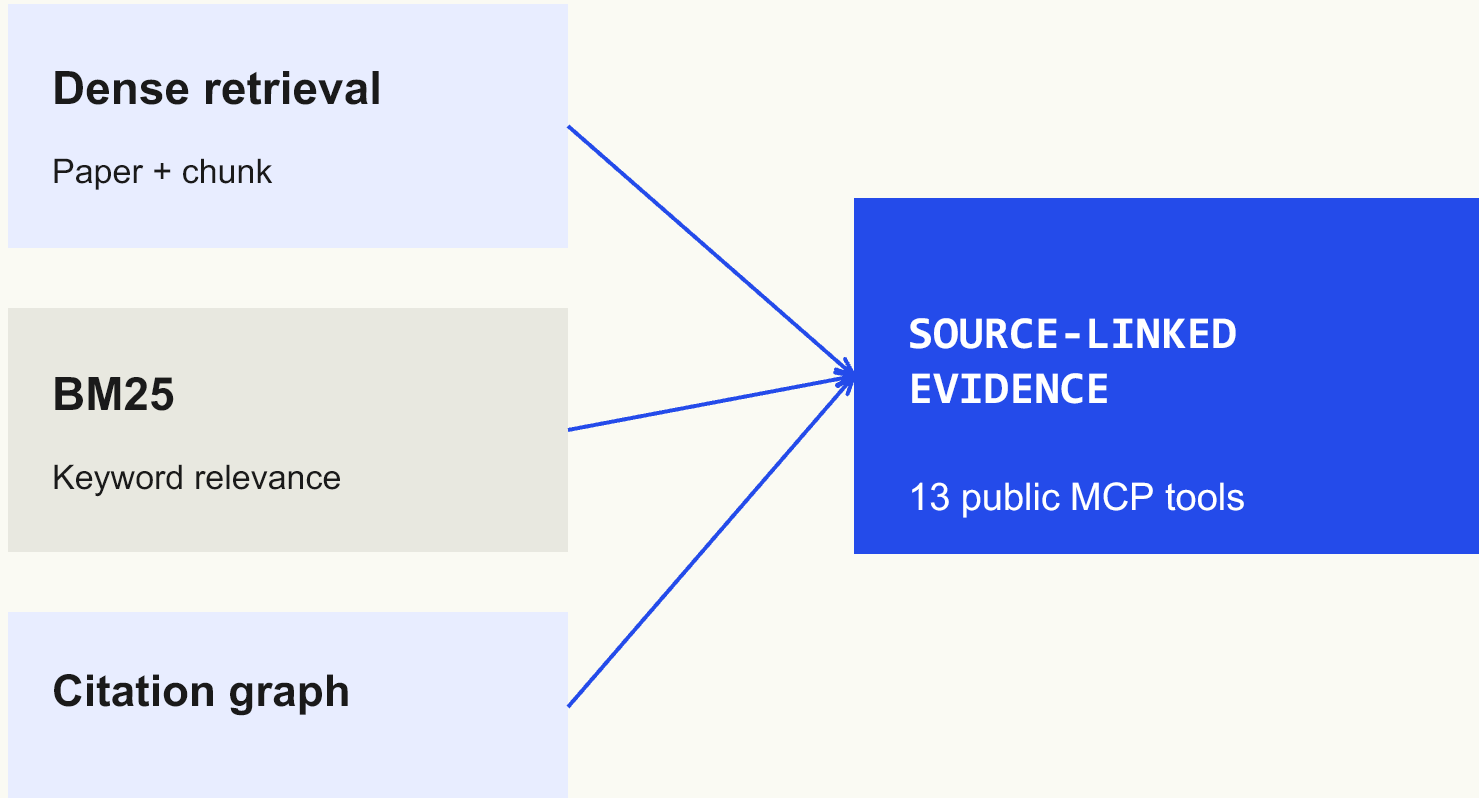


Research papers + protocol specs

SERVICE	FOCUS	WORKFLOW
dtox	Web3 papers + EIP/ERC/SIMD specs	Equations + structured evidence
Elicit	Literature synthesis + extraction	API + MCP
Semantic Scholar	Discovery + metadata	Citation graph
arxiv-mcp	arXiv papers	Original LaTeX + topic watches

Web3 papers + protocol specs in one research workflow.

Research you can inspect



17,365

Web3 research records

180,834

indexed documents

Coverage varies by record.

Free today. Paid depth is next.

LIVE

Free public research access

The public read endpoint works without a wallet.

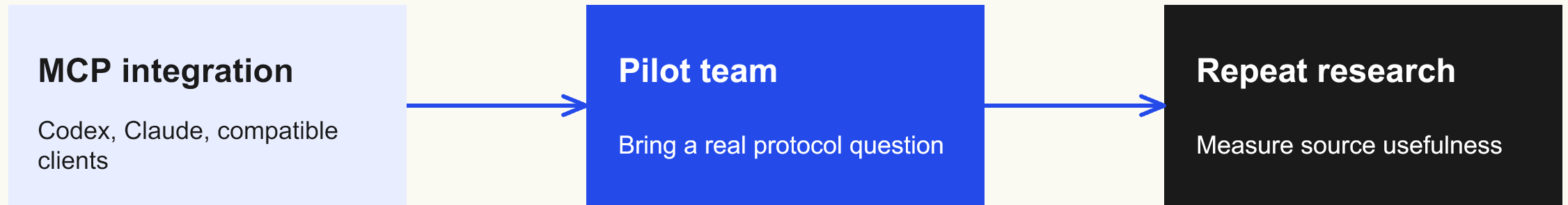
PROPOSED

Usage-based access for intensive research

Pilot willingness to pay before setting prices.

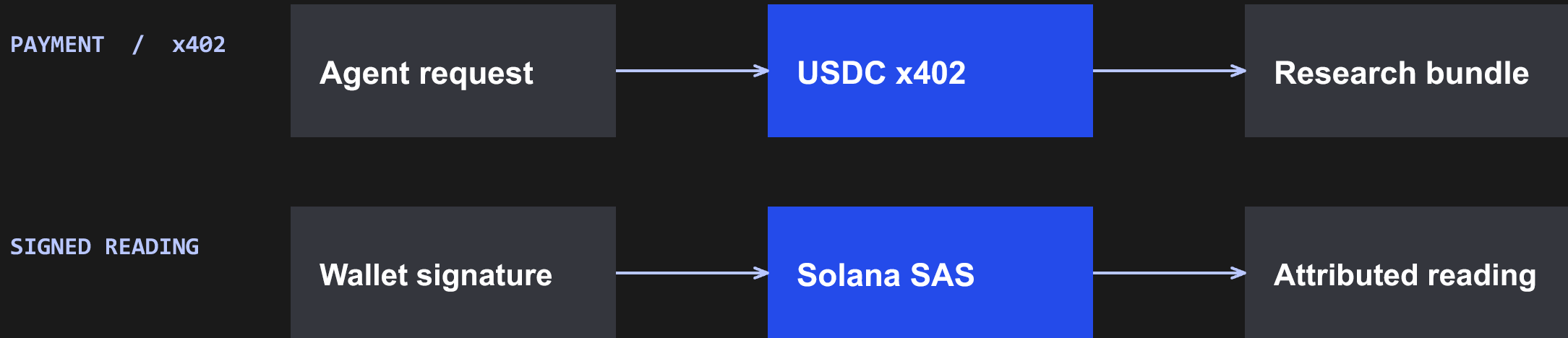
First test: does deeper research earn repeat use?

Meet builders inside their agents



Pilot target: useful sources and repeat use.

Solana powers payments and signed research records



DEVNET DEMONSTRATED

Base, Ethereum, Arbitrum Sepolia configured; E2E pending

These chain features are optional. Research remains free to read.

Signature proves attribution, not truth.

FOUNDER

Daniyar Gabdullin

Solo founder

MOBILE PRODUCT ENGINEERING

Build and ship customer-facing products

WEB3 PRODUCT QA

Test wallets, dApps, and blockchain flows

[Portfolio](#)

[LinkedIn: Daniyar Gabdullin](#)

PILOT INVITATION

Bring one real Web3 question.

PROPOSED PILOT TARGET

3 teams

Measure source usefulness
and repeat research.

read.whoim.space/research/

github.com/who1900/dtox-research